



2015 HSC Course Assessment Task 3 (Trial Examination)
June 18, 2015

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 13)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

- ☐ 12M2A – Miss Lee
 ☐ 12M3A – Mrs Ziaziaris
☐ 12M3B – Mr Berry
☐ 12M3C – Mr Zuber

Marker's use only.

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. Solving the equation $2^{2x} - 5(2^x) + 4 = 0$ gives 2 solutions for x . Which pair of solutions is correct? 1

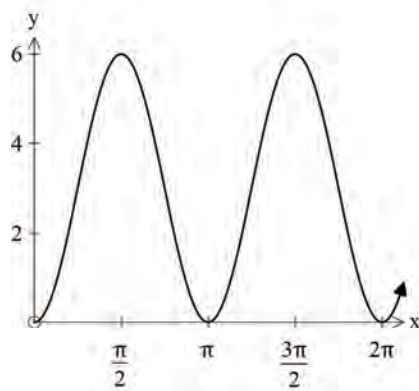
(A) $x = 1$ or $x = 0$

(C) $x = 2$ or $x = 0$

(B) $x = \log_2 2$ or $x = \log_2 1$

(D) $x = 4$ or $x = 1$

2. The equation of the graph below is given by $y = A \cos Bx + 3$. 1



Which of the following are the values of A and B ?

(A) $A = 3, B = 2$

(C) $A = 6, B = 3$

(B) $A = -3, B = 2$

(D) $A = -6, B = \pi$

3. A parabola has its focus at $(0, 2)$. The equation of its directrix is $x = -2$. Which of the following is the equation of the parabola? 1

(A) $(x + 1)^2 = 4(y - 2)$

(C) $(y + 1)^2 = 4(x - 2)$

(B) $x^2 = 8y$

(D) $y - 2)^2 = 4(x + 1)$

4. The value of $\log_3 5000$ is closest to: 1

(A) 1.5

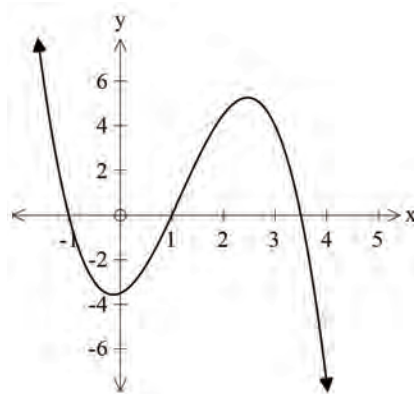
(C) 2.2

(B) 7.8

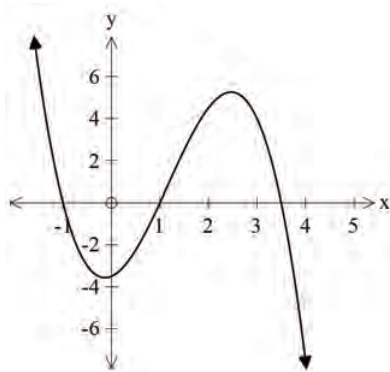
(D) 5.7

5. Examine the graph of $f(x)$ supplied.

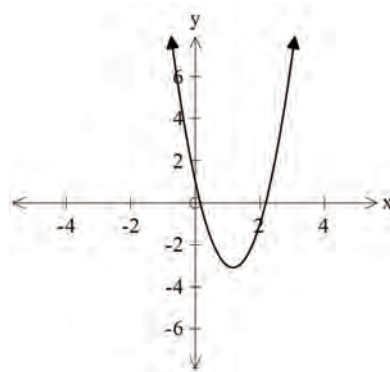
1



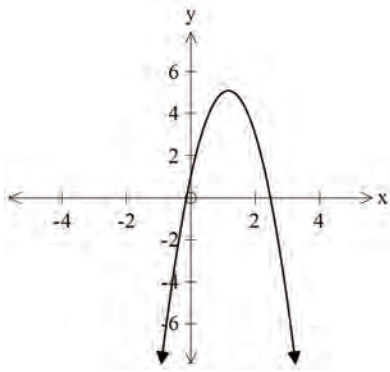
Which of the graphs below best represents $f'(x)$?



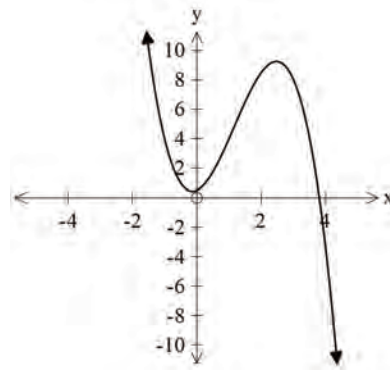
(A)



(C)



(B)



(D)

6. A circle has the equation $x^2 - 8x + y^2 - 1 = 0$. It has a radius of:

1

(A) 17

(C) 1

(B) 4

(D) $\sqrt{17}$

7. If $\sin \theta = \frac{5}{13}$ and $\cos \theta < 0$, what is the exact value of $\tan \theta$?

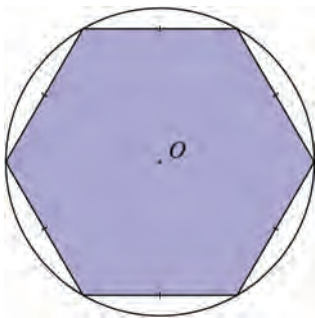
1

(A) $\frac{5}{12}$ (C) $-\frac{5}{12}$ (B) $\frac{12}{5}$ (D) $-\frac{12}{5}$

8. Suppose that the point $P(a, f(x))$ lies on the curve $y = f(x)$. 1
If $f'(a) = 0$ and $f''(a) > 0$, which of the following statements describes the point P on the graph of $y = f(x)$?

- (A) P is a maximum turning point (C) P is a stationary point of inflexion
(B) P is a minimum turning point (D) P is a point of inflexion

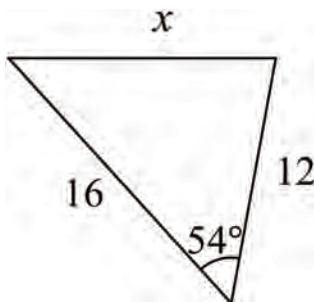
9. A regular hexagon is cut from a circle with centre O , such that each vertex of the hexagon lies on the circumference of the circle. 1
The hexagon lies on the circumference of the circle.



What percentage of the circle (to the nearest whole number) is the area of the hexagon?

- (A) 85% (C) 83%
(B) 84% (D) 82%

10. The value of x in the diagram to the nearest whole number is: 1



- (A) 13 (C) 25
(B) 153 (D) 10

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Solve $ 2x - 1 < 5$		2
(b) Fully factorise $2x^3 - 54$		2
(c) Simplify fully $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta}$		2
(d) State the domain of $f(x) = \sqrt{81 - x^2}$		1
(e) Rationalise the denominator and simplify $\frac{1 - \sqrt{2}}{3 + \sqrt{2}}$		2
(f) The first term of an arithmetic series is 3, and the ninth term is five times the second term. Find the common difference.		2
(g) Differentiate the following with respect to x . Simplify where possible.		
i. $\sqrt{1 - 2x}$		2
ii. $\frac{e^{2x}}{x^2}$		2

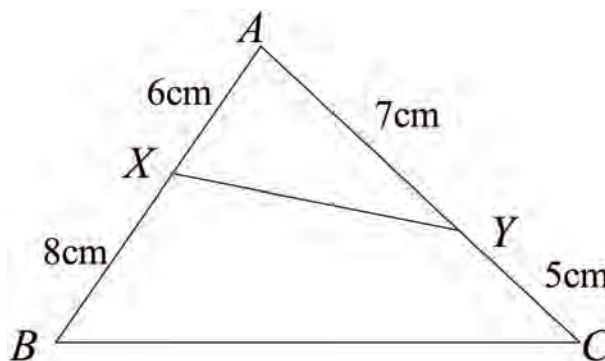
Question 12 (15 Marks)	Commence a NEW page.	Marks
(a) Evaluate the following integrals:		
i. $\int (\sec^2 x + 3 \cos x) dx$		2
ii. $\int 2^x dx$		2
(b) Evaluate $\int_2^7 \frac{x}{x^2 - 1} dx$, leaving your answer in the simplest exact form.		3
(c) For the equation $2x^2 + 3x - 7 = 0$, evaluate:		
i. $\alpha + \beta$		1
ii. $\alpha\beta$		1
iii. $(\alpha - 2)(\beta - 2)$		2
(d) A function is defined by $f(x) = 2x^3 - 6x + 3$.		
i. Find the coordinates of the turning points of the graph $y = f(x)$ and determine their nature.		2
ii. Hence sketch the graph of $y = f(x)$, showing the turning points and the y intercept.		2

Question 13 (15 Marks)

Commence a NEW page.

Marks

- (a) The points A and B have coordinates $(1, 0)$ and $(7, 4)$ respectively. The angle between the line AB and the x -axis is θ .
- Find the gradient of the line AB . 1
 - Calculate the size of angle θ in degrees. 1
 - Find the length of interval AB . 1
 - Find the equation of the line AB . 2
 - Find the coordinates of C , the midpoint of AB . 1
- (b) Copy the diagram below into your examination booklet.



- Prove $\triangle ABC \parallel \triangle AXY$. 2
 - If $XY = 9$ cm, find the length of BC . 1
- (c) For what value(s) of k does the equation below have no real roots? 3

$$4x^2 + (k + 3)x + 1 = 0$$

- (d) Using Simpson's rule with five function values, find an estimate for the definite integral, correct to 2 decimal places: 3

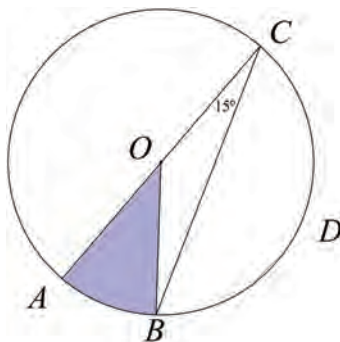
$$\int_2^6 \log_e(x - 1) dx$$

Question 14 (15 Marks)

Commence a NEW page.

Marks

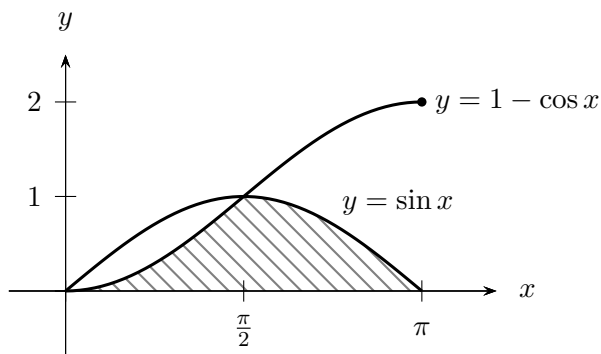
- (a) In the figure, O is the centre of the circle. The length of minor arc $AB = 5\text{cm}$, $\angle ACB = 15^\circ$.

Find in terms of π :

- i. the length of radius OA 2
 - ii. the area of the shaded sector AOB 1
 - iii. the area of the minor segment BDC 2
- (b) The diagram shows the graphs of the functions $y = 1 - \cos x$ and $y = \sin x$ between $x = 0$ and $x = \pi$. 4

The graphs intersect at $x = \frac{\pi}{2}$.

Find the area of the shaded region.



- (c) For the geometric series

$$1 - 3x + 9x^2 - \dots$$

- i. find the values of x for which the limiting sum exists. 2
 - ii. find the value of x for which the limiting sum is $\frac{4}{5}$ 2
- (d) Shade the region which satisfies all the inequalities: 2

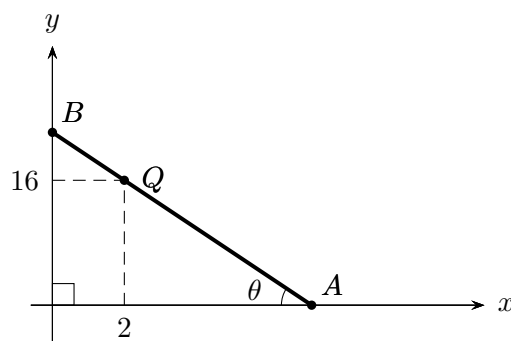
$$\begin{cases} x \geq 0 \\ y \geq x^2 \\ y \leq \sqrt{9 - x^2} \end{cases}$$

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) Evaluate $\sum_{k=2}^5 \frac{k^2}{k+1}$ **2**
- (b) Find the equation of the locus of the point $P(x, y)$ which moves so that it is equidistant from $A(-1, 2)$ and $B(5, -3)$. **2**
- (c) R is the region bounded by the y -axis, the x -axis, the line $x = \frac{\pi}{2}$ and the curve $y = \sqrt{1 + \sin x}$. Show that the volume of the solid formed when R is rotated about the x -axis is **3**
- $$V = \frac{1}{2}\pi(\pi + 2)$$
- (d) i. Prove that the x -values of the points of intersection of the hyperbola $y = \frac{k}{x}$ and the line $kx + y + 2 = 0$ are given by the solution of the equation $kx^2 + 2x + k = 0$. **1**
- ii. Find the values of k for which the hyperbola and the line will intersect in two distinct points. **2**
- (e) A straight road is to be built from A to B . The road must pass through Q , a vertex of the rectangular block of land 16 km by 2 km as shown in the diagram below. AB makes an angle of θ with AP . **1**



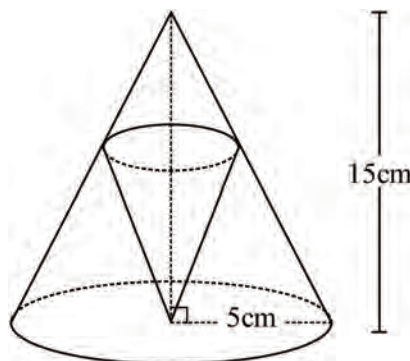
- i. Show $AB = \frac{16}{\sin \theta} + \frac{2}{\cos \theta}$ **1**
- ii. Show that $\tan \theta = 2$ gives the minimum distance for AB **4**

Question 16 (15 Marks)

Commence a NEW page.

Marks

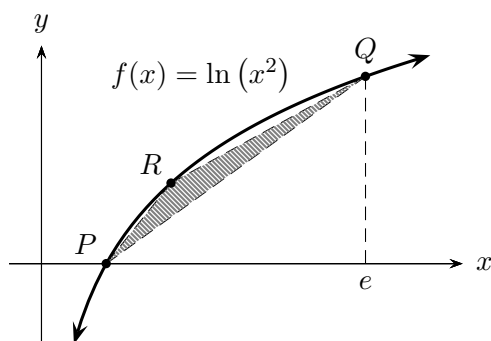
- (a) A small cone is enclosed within a larger cone as shown in the diagram below. The large cone has height 15cm and radius 5cm. Let h represent the height of the small cone and r represent the radius.



- i. Show $h = 15 - 3r$ **1**
 - ii. Find the dimensions of the small cone for which the volume of the small cone is maximum. **3**
- (b) The diagram shows the graph of the function $y = \ln(x^2)$, ($x > 0$).

The points $P(1, 0)$, $Q(e, 2)$ and $R(t, \ln t^2)$ all lie on the curve.

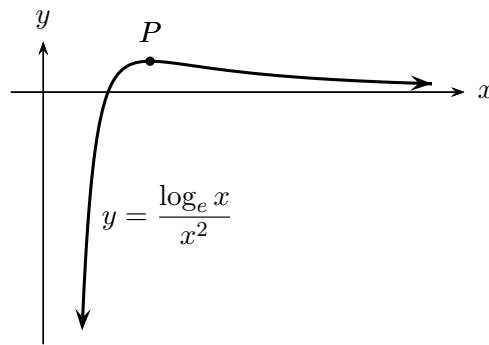
The area of $\triangle PQR$ is maximum when the tangent at R is parallel to the line through P and Q .



- i. Find gradient of the line through P and Q . **1**
- ii. Find the value of t that gives the maximum area for $\triangle PQR$. **2**
- iii. Hence find the maximum area of $\triangle PQR$ **2**

Examination continues overleaf...

- (c) The diagram below is the graph of $y = \frac{\log_e x}{x^2}$. P is a maximum turning point.



- i. Find the coordinates of the point P . **3**
- ii. Find the values of h such that $\frac{\log_e x}{x^2} = h$ has two values. **1**
- iii. Find the values of k such that $\frac{\log_e x}{x^2} = kx$ has two solutions. **2**

End of paper.

2 unit Trial 2015

1. $a^2 - 5a + 4 = 0$

$(a-4)(a-1) = 0$

$a = 4, a = 1$

$2^x = 4, 2^x = 1$

$x = 2, x = 0$

C

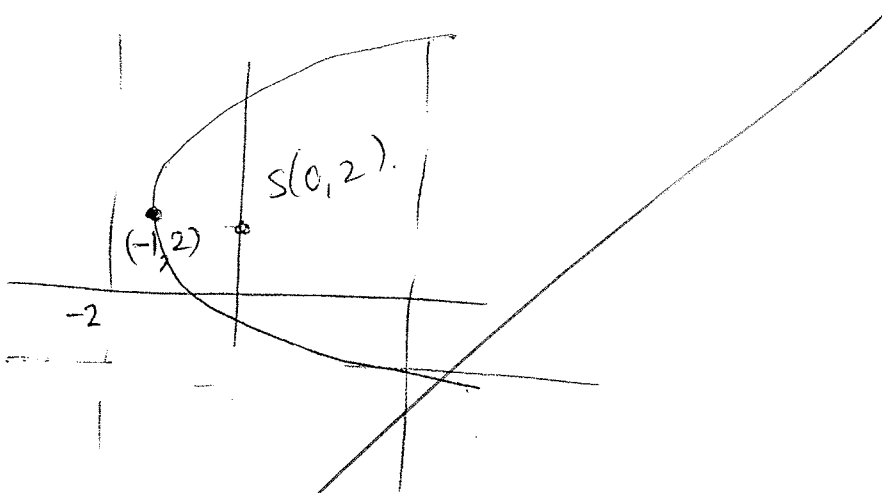
2. $\frac{2\pi}{10} = \pi$

$\therefore b = 2$

$a = -3$

B

3.



Vertex $(-1, 2)$ $a = 1$

$(y-2)^2 = 4a(x+1)$

$(y-2)^2 = 4(x+1)$

D.

4. $\frac{\log_e 5000}{\log_e 3}$

B

5.

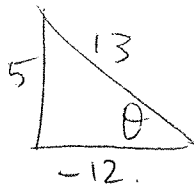
B

6. $x^2 - 8x + (-4)^2 + y^2 = 1 + 16$

$(x-4)^2 + y^2 = 17$

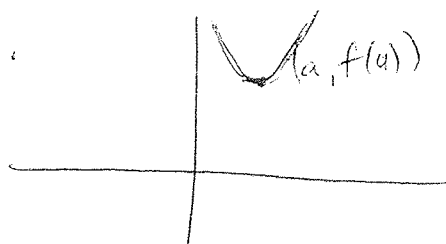
D.

7.



C

8.



B

$$A_0 = \pi r^2$$

$$A_{\text{hex}} = \frac{1}{2} \times r^2 \sin 60^\circ \times 6$$

$$= \frac{r^2}{2} \times \frac{\sqrt{3}}{2} \times 6$$

$$= \frac{r^2 \sqrt{3}}{4} \times 6$$

$$= \frac{3\sqrt{3}r^2}{2}$$

$$\frac{3\sqrt{3}r^2}{2} \div \pi r^2 \times 100\%$$

$$= \frac{3\sqrt{3}}{2} \times \frac{1}{\pi}$$

$$= \frac{3\sqrt{3}}{2\pi} \times 100\%$$

$$= 83\%$$

C

$$10. \quad x = \sqrt{16^2 + 12^2 - 2(16)(12)\cos 54}$$

$$= 13.2 \quad (1 \text{ d.p.})$$

A

Section II

-1 if not in order

11. a) $|2x-1| < 5$

$$-5 < 2x-1 < 5$$

$$-4 < 2x < 6$$

$$\boxed{-2 < x < 3}$$

b) $2x^3 - 54$

$$= 2(x^3 - 27)$$

$$= 2(x-3)(x^2 + 3x + 9)$$

c) $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta}$

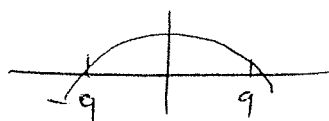
$$= \frac{\sin^2 \theta}{\sin \theta \cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta$$

d) D: $81 - x^2 \geq 0$

$$(9-x)(9+x) \geq 0$$



$$\boxed{-9 \leq x \leq 9}$$

e) $\frac{(1 - \sqrt{2}) \times (3 - \sqrt{2})}{(3 + \sqrt{2})(3 - \sqrt{2})}$

$$= \frac{3 - \sqrt{2} - 3\sqrt{2} + 2}{9 - 2}$$

$$= \frac{5 - 4\sqrt{2}}{7}$$

$$= \frac{5 - 4\sqrt{2}}{7}$$

7

f) $a = 3$

$$a + 8d = 5(a + d)$$

$$3 + 8d = 5(3 + d)$$

$$3 + 8d = 15 + 5d$$

$$3d = 12$$

$$d = 4$$

g) (i) $\frac{d}{dx} (1-2x)^{1/2}$

$$= \frac{1}{2}(-2)(1-2x)^{-1/2}$$

$$= -\frac{1}{\sqrt{1-2x}}$$

Need to simplify

(ii) $\frac{d}{dx} \left(\frac{\sin 2x}{x^2} \right)$

$$= \frac{x^2 \cdot 2 \cos 2x - \sin 2x \cdot 2x}{x^4}$$

$$= \frac{2x(x \cos 2x - \sin 2x)}{x^4}$$

$$= \frac{2(x \cos 2x - \sin 2x)}{x^3}$$

Don't need to simplify

$$\frac{d}{dx} \left(\frac{e^{2x}}{x^2} \right)$$

$$= \frac{2e^{2x}x^2 - 2xe^{2x}}{x^4}$$

$$= \frac{2xe^{2x}(x-1)}{x^4}$$

$$= \frac{2e^{2x}(x-1)}{x^3}$$

$$12. a) (i) \int (\sec^2 x + 3 \cos x) dx$$

$$= \tan x + 3 \sin x + C. \quad 2$$

$$(ii) \int 2^x dx$$

$$= \int e^{\ln 2^x} dx$$

$$= \int e^{x \ln 2} dx$$

$$= \frac{e^{x \ln 2}}{\ln 2} \quad 1$$

$$= \frac{2^x}{\ln 2} + C \quad 1$$

$$b) \int_2^7 \frac{x}{x^2-1} dx$$

$$= \frac{1}{2} \int_2^7 \frac{2x}{x^2-1} dx$$

$$= \frac{1}{2} [\ln |x^2-1|]_2^7 \quad 1$$

$$= \frac{1}{2} [\ln 48 - \ln 3]$$

$$= \frac{1}{2} \ln \left(\frac{48}{3} \right) \quad 1$$

$$= \frac{1}{2} \ln 16$$

$$= \ln 16^{\frac{1}{2}}$$

$$= \ln 4. \quad 1$$

$$c) (i) 2x^2 + 3x - 7 = 0$$

$$\alpha + \beta = -\frac{3}{2} \quad 1$$

$$(ii) \alpha \beta = -\frac{7}{2} \quad 1$$

$$(iii) (\alpha-2)(\beta-2)$$

$$= \alpha\beta - 2\alpha - 2\beta + 4$$

$$= \alpha\beta - 2(\alpha+\beta) + 4 \quad 1$$

$$= -\frac{7}{2} - 2\left(-\frac{3}{2}\right) + 4$$

$$= -\frac{7}{2} + 3 + 4$$

$$= 3\frac{1}{2} \quad 1$$

$$d) f(x) = 2x^3 - 6x + 3$$

$$(i) f'(x) = 6x^2 - 6$$

$$f''(x) = 12x$$

$$\text{Stat Pts: } f'(x) = 0$$

$$6x^2 - 6 = 0$$

$$6x^2 = 6$$

$$x^2 = 1$$

$$x = 1, -1$$

$$y = -1, 7$$

$$\text{Test } (1, -1)$$

$$f''(x) = 12$$

$$> 0$$

$$\therefore \boxed{\text{min}(1, -1)} \quad 1$$

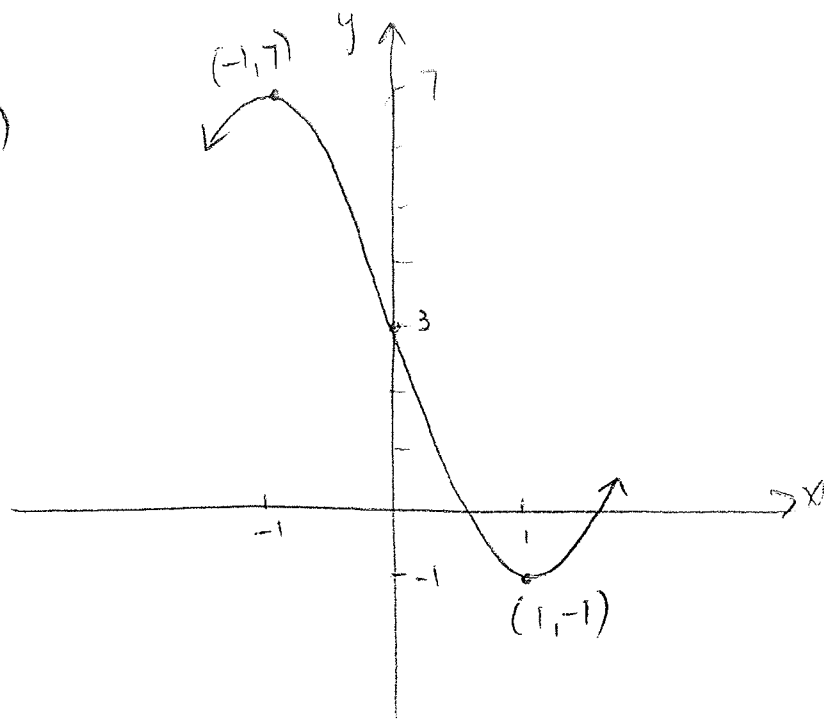
$$\text{Test } (-1, 7)$$

$$f''(x) = -12$$

$$< 0$$

$$\therefore \boxed{\text{max } -1, 7} \quad 1$$

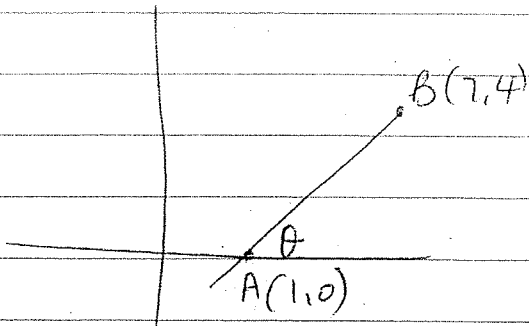
(ii)



1 y-intercept

1 turning pts

13. a) A(1,0) B(7,4)



$$(i) m = \frac{4}{6} = \frac{2}{3}$$

$$(ii) \tan \theta = \frac{2}{3}$$

$$\theta = 33^{\circ} 41'$$

$$\begin{aligned} (iii) d &= \sqrt{(7-1)^2 + (4-0)^2} \\ &= \sqrt{36 + 16} \\ &= \sqrt{52} \text{ units} \\ &= 7.2 \text{ units (1dp)} \end{aligned}$$

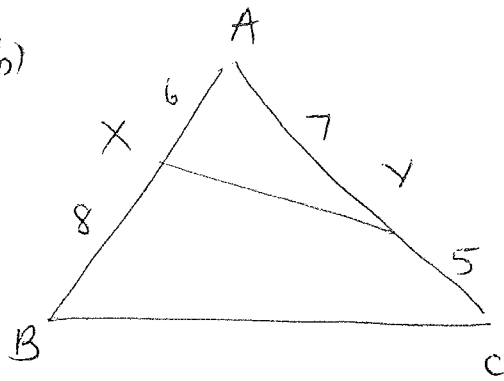
$$(iv) y - 0 = \frac{2}{3}(x - 1)$$

$$3y = 2x - 2$$

$$2x - 3y - 2 = 0$$

$$\begin{aligned} (v) C(x,y) &= \left(\frac{1+7}{2}, \frac{4+0}{2} \right) \\ &= (4, 2) \end{aligned}$$

13. b)



(i) In Δ 's AXY & ACB

1. $\angle A$ is common

$$2. \frac{AY}{AB} = \frac{7}{14} = \frac{1}{2}$$

$$\frac{AX}{AC} = \frac{6}{12} = \frac{1}{2}$$

$\therefore \Delta AXY \parallel \Delta ACB$

(two sides in proportion
a included angle equal)

(ii) $BC = 18$ cm

(corresponding sides of
similar triangles in
same ratio)

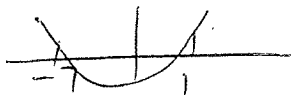
c) No real roots: $b^2 - 4ac < 0$

$$(k+3)^2 - 4(4)(1) < 0$$

$$k^2 + 6k + 9 - 16 < 0$$

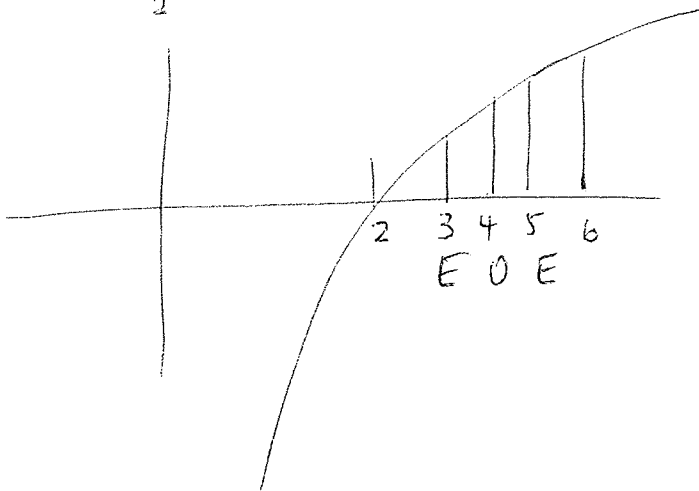
$$k^2 + 6k - 7 < 0$$

$$(k+7)(k-1) < 0$$



$$\boxed{-7 < k < 1}$$

d) $\int_2^6 \log_e(x-1) dx$



$$A \doteq \frac{1}{3} \{ f(2) + 4[f(3) + f(5)] + 2[f(4)] + f(6) \} \quad |$$

$$= \frac{1}{3} \{ \ln 1 + 4(\ln 2 + \ln 4) + 2 \ln 3 + \ln 5 \} \quad |$$

$$= 4.04 \quad |$$

14. a) (i) $\angle BOC = 15^\circ$ (isos. Δ)
 $\therefore \angle AOB = 30^\circ$ (ext. angle of a triangle)
 $= \frac{\pi}{6}$ rule

$$l = r\theta$$

$$5 = r \times \frac{\pi}{6}$$

$$r = \frac{30}{\pi} \text{ cm.}$$

(ii) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times \left(\frac{30}{\pi}\right)^2 \times \frac{\pi}{6}$
 $= \frac{900\pi}{12\pi^2}$
 $= \frac{75}{\pi} \text{ cm}^2.$

(iii) $\angle COB = 150^\circ = \frac{150 \times \pi}{180}$
 $= \frac{5\pi}{6}$

$$A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times \left(\frac{30}{\pi}\right)^2 \left(\frac{5\pi}{6} - \sin \frac{5\pi}{6}\right)$$

$$= \frac{1}{2} \times \frac{900}{\pi^2} \left(\frac{5\pi}{6} - \frac{1}{2}\right)$$

$$= \frac{450}{\pi^2} \times \frac{5\pi}{6} - \frac{225}{\pi^2}$$

$$= \frac{375}{\pi} - \frac{225}{\pi^2}$$

$$= \frac{375\pi - 225}{\pi^2}$$

$$= 96.57 \text{ cm}^2 \text{ (2 dp)}$$

$$\begin{aligned}
 b) \quad A &= \int_0^{\pi/2} (1 - \cos x) dx + \int_{\pi/2}^{\pi} \sin x dx & 1 \\
 &= \left[x - \sin x \right]_0^{\pi/2} + \left[-\cos x \right]_{\pi/2}^{\pi} & 1 \\
 &= \frac{\pi}{2} - \sin \frac{\pi}{2} + \left[-\cos \pi + \cos \frac{\pi}{2} \right] & 1 \\
 &= \frac{\pi}{2} - 1 + 1 + 0 \\
 &= \frac{\pi}{2} \quad u^2 & 1
 \end{aligned}$$

c) (i) Limiting sum exists when

$$\begin{aligned}
 -1 &< r < 1 \\
 -1 &< -3x < 1 \\
 1 &> x > -\frac{1}{3}
 \end{aligned}$$

$$\therefore \boxed{-\frac{1}{3} < x < \frac{1}{3}}$$

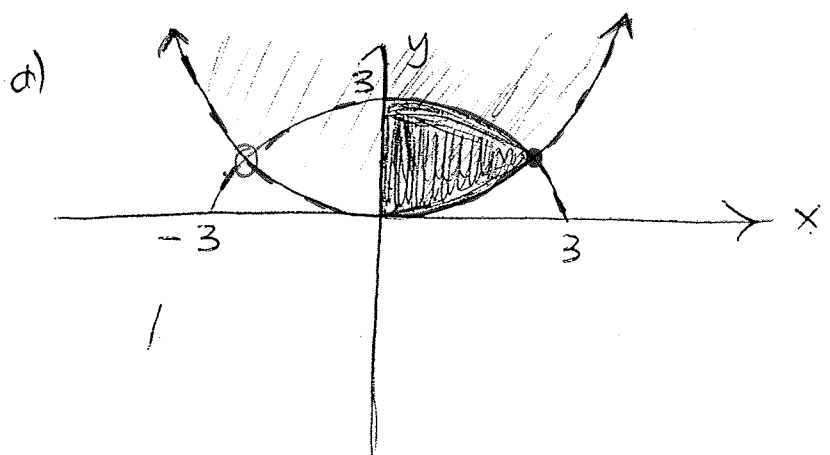
$$(ii) \quad S_{\infty} = \frac{a}{1-r}$$

$$\frac{4}{5} = \frac{1}{1+3x}$$

$$4 + 12x = 5$$

$$12x = 1$$

$$x = \frac{1}{12}$$



1 curves

1 region

$$15. (a) \sum_{k=2}^5 \frac{k^2}{k+1}$$

$$k=2: \frac{2^2}{3} = \frac{4}{3}$$

$$k=3: \frac{3^2}{4} = \frac{9}{4}$$

$$k=4: \frac{4^2}{5} = \frac{16}{5}$$

$$k=5: \frac{5^2}{6} = \frac{25}{6} \quad |$$

$$\text{Sum} = \frac{219}{20} \quad |$$

$$(b) (x+1)^2 + (y-2)^2 = (x-5)^2 + (y+3)^2 \quad |$$

$$x^2 + 2x + y^2 - 4y + 5 = x^2 - 10x + y^2 + 6y + 34$$

$$12x - 10y - 29 = 0 \quad |$$

$$c) V = \pi \int_0^{\pi/2} (1 + \sin x) dx \quad |$$

$$= \pi \int_0^{\pi/2} (1 + \sin x) dx \quad |$$

$$= \pi \left[\frac{\pi}{2} - \cos \frac{\pi}{2} - 0 + 1 \right]$$

$$= \pi \left[\frac{\pi}{2} + 1 \right]$$

$$V = \frac{1}{2} \pi [\pi + 2] \quad |$$

15 d) (i) Solve simultaneously

$$y = \frac{k}{x} \quad \text{--- (1)}$$

$$kx + y + 2 = 0 \quad \text{--- (2)}$$

Sub (1) into (2)

$$kx + \frac{k}{x} + 2 = 0$$

$$kx^2 + k + 2x = 0$$

(ii) 2 distinct points $\Delta > 0$.

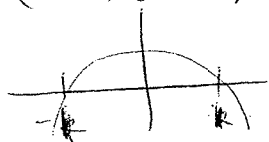
$$b^2 - 4ac > 0$$

$$4 - 4(k)(k) > 0$$

$$4 - 4k^2 > 0$$

$$1 - k^2 > 0$$

$$(1-k)(1+k) > 0$$



$$\boxed{-1 < k < 1}$$

e) (i) $\sin \theta = \frac{16}{AQ}$

$$AQ = \frac{16}{\sin \theta}$$

$\angle BQN = \theta$ (corresponding angles on parallel lines)

$$\therefore \cos \theta = \frac{2}{BQ}$$

$$BQ = \frac{2}{\cos \theta}$$

$$AB = AQ + BQ$$

$$\therefore AB = \frac{16}{\sin \theta} + \frac{2}{\cos \theta}$$

$$\text{iii) } AB = 16(\sin \theta)^{-1} + 2(\cos \theta)^{-1}$$

$$(AB)' = -16(\sin \theta)^{-2} \cos \theta - 2(\cos \theta)^{-2} (-\sin \theta)$$

$$= \frac{-16 \cos \theta}{\sin^2 \theta} + \frac{2 \sin \theta}{\cos^2 \theta}$$

$$\text{Stat Pts: } (AB)' = 0.$$

$$\frac{-16 \cos \theta}{\sin^2 \theta} + \frac{2 \sin \theta}{\cos^2 \theta} = 0.$$

$$\frac{-16 \cos^3 \theta + 2 \sin^3 \theta}{\sin^2 \theta \cos^2 \theta} = 0.$$

$$-2(8 \cos^3 \theta - \sin^3 \theta) = 0.$$

$$8 \cos^3 \theta - \sin^3 \theta = 0$$

$$(2 \cos \theta - \sin \theta)(4 \cos^2 \theta + 2 \cos \theta \sin \theta + \sin^2 \theta) = 0$$

$$2 \cos \theta - \sin \theta = 0.$$

$$2 \cos \theta = \sin \theta.$$

$$2 = \tan \theta.$$

$$\text{Test } \tan \theta = 2$$

	1	$\tan^{-1} 2$	2
$(AB)'$	-6	0	19

$\therefore \swarrow$

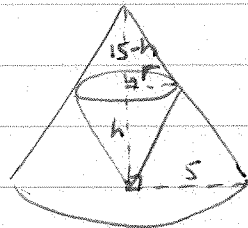
$\therefore$ minimum when $\tan \theta = 2$.

16. a) i) $\frac{15-h}{r} = \frac{15}{5}$ (corresp. sides of similar triangles in proportion)

$$\frac{15-h}{r} = 3$$

$$15-h = 3r$$

$$h = 15-3r$$



ii) $V = \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3}\pi r^2 (15-3r)$$

$$= 5\pi r^2 - \pi r^3$$

$$V' = 10\pi r - 3\pi r^2$$

$$V'' = 10\pi - 6\pi r$$

Stat. Pts : $V' = 0$

$$10\pi r - 3\pi r^2 = 0$$

$$\pi r (10-3r) = 0$$

$$r > 0, r = \frac{10}{3}$$

But $r > 0$.

Test $r = \frac{10}{3}$

$$V'' = 10\pi - 6\pi r$$

$$= 10\pi - 6\pi \left(\frac{10}{3}\right)$$

$$< 0$$

$\therefore$ Max at $r = \frac{10}{3}$

and $h = 15 - 3\left(\frac{10}{3}\right)$

$h = 5$

$$16 \text{ b) (i) } m_{PQ} = \frac{2-0}{e-1}$$

$$= \frac{2}{e-1}$$

$$(ii) y = \log_e x^2 = 2 \log_e x$$

$$\frac{dy}{dx} = 2 \times \frac{1}{x} = \frac{2}{x}$$

$$\text{Now } m = m_{PQ}$$

$$\therefore \frac{2}{x} = \frac{2}{e-1}$$

$$2(e-1) = 2x$$

$$x = e-1$$

$$\therefore x = e-1$$

$$(iii) R(e-1, 2\ln(e-1))$$

$$d_{PQ} = \sqrt{(e-1)^2 + 2^2}$$

$$= \sqrt{e^2 - 2e + 1 + 4}$$

$$= \sqrt{e^2 - 2e + 5}$$

$$\text{Eqn}_{PQ} \frac{y-0}{e-1} = \frac{2}{e-1} (x-1)$$

$$y = \frac{2}{e-1} (x-1)$$

$$(e-1)y = 2x - 2$$

$$2x - (e-1)y - 2 = 0$$

$$\text{Perp dist} = \frac{|2(e-1) - (e-1)2\ln(e-1) - 2|}{\sqrt{4 + (e-1)^2}}$$

$$= \frac{|2e - 2 - 2(e-1)\ln(e-1) - 2|}{\sqrt{4 + e^2 - 2e + 1}}$$

$$= \frac{|2e - 4 - 2(e-1)\ln(e-1)|}{\sqrt{5 + e^2 - 2e}}$$

$$A_{\Delta PQR} = \frac{1}{2} \times \sqrt{e^2 - 2e + 5}$$

$$\times \frac{|2e - 4 - 2(e-1)\ln(e-1)|}{\sqrt{e^2 - 2e + 5}}$$

$$= \frac{2e - 4 - 2(e-1)\ln(e-1)}{2}$$

$$= [e - 2 - (e-1)\ln(e-1)]$$

16(a)

$$(1) y = \frac{\log_e x}{x^2}$$

$$\frac{dy}{dx} = \frac{x^2 \cdot \frac{1}{x} - \log_e x \cdot 2x}{x^4}$$

$$= \frac{x - 2x \ln x}{x^4} = \frac{1 - 2 \ln x}{x^3} \quad |$$

$$\text{Stat Pts: } \frac{dy}{dx} = 0$$

$$\frac{x - 2x \ln x}{x^4} = 0$$

$$| \frac{1 - 2 \ln x}{x^3} = 0$$

$$x(1 - 2 \ln x) = 0$$

$$1 - 2 \ln x = 0$$

$$x = 0, \quad 2 \ln x = 1$$

$$2 \ln x = 1$$

$$\ln x = \frac{1}{2}$$

$$\ln x = \frac{1}{2}$$

$$e^{1/2} = x$$

$$y = \frac{\ln e^{1/2}}{(e^{1/2})^2}$$

$$y = \frac{\frac{1}{2}}{e^1}$$

$$= \frac{1}{2e}$$

$$\therefore P(e^{1/2}, \frac{1}{2e}) \quad |$$

(ii) As $x \rightarrow \infty$ $y \rightarrow 0$

$\frac{\log_e x}{x^2} = h$ has two values when

$$h > 0 \text{ but } h < \frac{1}{2e}$$

$$\text{i.e. } 0 < h < \frac{1}{2e}$$

(iii) $k = \frac{\ln x}{x^3}$

k max/min when $k' = 0$

$$k' = \frac{\frac{1}{x} \cdot x^3 - 3x^2 \cdot \ln x}{x^6}$$

$$= \frac{1 - 3 \ln x}{x^4}$$

$$= x^{-4} - 3x^{-4} \ln x$$

$$k'' = -4x^{-5} - (-12x^{-5} \ln x + \frac{1}{x} \cdot 3x^{-4})$$

$$= \frac{-4 + 12 \ln x - 3}{x^5}$$

$$k' = 0$$

$$1 - 3 \ln x = 0$$

$$3 \ln x = 1$$

$$x = e^{\frac{1}{3}}$$

$$k'' = \frac{-4 + 12 \times \frac{1}{3} - 3}{e^{\frac{5}{3}}}$$

$$< 0$$

$$k = \frac{\ln e^{\frac{1}{3}}}{(e^{\frac{1}{3}})^3}$$

$$= \frac{1}{3e}$$

$\frac{\ln x}{x^2} = k$ has two values when

$$k > 0 \text{ but } k < \frac{1}{3e}$$

i.e.

$$0 < k < \frac{1}{3e}$$